



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Ethernet and MAC Address Learning

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Duration: 3 months

Assignment Submission Date: Friday, 11th September 2026

Day:5 (Five)

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1. Question and Answers

1. What is Ethernet?

- Ethernet is a common networking technology used to connect devices in a wired LAN. It allows computers and other devices to communicate and share data over a network.

2. What is an Ethernet frame?

- An Ethernet frame is a unit of data sent between devices over an Ethernet network. It contains information such as the source MAC address, destination MAC address, actual data and error-checking information.

3. What is a MAC address?

- A MAC address is a unique address used to identify a network interface at the data link layer.

4. What does MAC stand for?

- MAC stands for Media Access Control.

5. Difference between a MAC address and an IP address?

- A MAC address identifies a network interface at Layer 2 and used mainly for local network communication. An IP address works at Layer 3 and used for logical addressing and routing between networks.

6. What is a MAC Address Table?

- A MAC address Table is a table maintained by a network switch that record MAC addresses and the switch ports where those devices are connected.

7. How does a switch learn MAC addresses?

- When a switch receives a frame, it examines the source MAC address. It then records that MAC address along with the port where the frame arrived.

8. What is a source MAC address?

- It identifies the interface that SENT the frame.

9. What is a destination MAC address?

- It identifies the intended receiver. A switch decides where to forward the frame.

10. What is an unknown unicast frame?

- An unknown unicast frame is a frame whose destination MAC address is not currently found in switch's MAC address Table.

11. What is flooding?

- Flooding is when switch sends an unknown unicast frame out of multiple other ports to know the correct destination port.

12. What is known unicast forwarding?
- Known unicast forwarding occurs when the switch already knows the destination MAC address. It checks the MAC address Table and sends frame only through appropriate port.
13. At which OSI layer does Ethernet switching primarily operate?
- Ethernet switching primarily operates at Layer 2 (Data Link Layer) of the OSI model.
14. Why does a switch maintain a MAC Address Table?
- A switch maintains a MAC Address Table so it can identify which device is connected to which port and forward frames efficiently instead of sending them to every port.
15. Why is Ethernet knowledge important in cybersecurity?
- Ethernet knowledge helps cybersecurity professionals understand local network traffic, MAC addresses, Ethernet frames, ARP switch behavior, and suspicious Layer 2 activity. It is also useful when working with tools such as Wireshark and Nmap during network security analysis.

2. PRACTICAL IDENTIFICATION (4 marks)

2.1 Record: Network adapter name, MAC address, IPv4 address, default gateway.

Network Adapter Name: Wi-Fi

MAC Address: 1C-CE-51-F4-2D-52

IPv4 Address: 192.168.101.5

Default Gateway: 192.168.101.1

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCI
e Adapter
Physical Address. . . . . : 1C-CE-51-F4-2D-52
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2404:7c00:48:6c03:dd5c:7b72:8ea2:de0d
(Preferred)
Temporary IPv6 Address. . . . . : 2404:7c00:48:6c03:64f1:c57a:a3da:eed3
(Preferred)
Link-local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5(Preferred
)
IPv4 Address. . . . . : 192.168.101.5(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, September 10, 2026 2:27:47
PM
Lease Expires . . . . . : Friday, September 11, 2026 2:27:46 PM
Default Gateway . . . . . : fe80::72a5:6aff:feb3:259a%5
192.168.101.1
DHCP Server . . . . . : 192.168.101.1
DHCPv6 IAID . . . . . : 85773905
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-8
0-C6
DNS Servers . . . . . : fe80::1%5
110.44.112.245
110.44.113.245
fe80::1%5
NetBIOS over Tcpip. . . . . : Enabled
```

2.2 SCREENSHOTS (Attach screenshots of: getmac, ipconfig /all, arp - a)

➤ getmac

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>getmac

Physical Address      Transport Name
=====
1C-CE-51-F4-2D-52     \Device\Tcpip_{224D9732-773F-423D-9543-56BC8C13281A}
30-13-8B-85-80-C6     Media disconnected
1C-CE-51-F4-2D-53     Media disconnected
0A-00-27-00-00-0A     \Device\Tcpip_{3BE25071-B5CA-4DFC-8956-4D9BDFE82737}
```

➤ ipconfig /all

```
C:\Users\HP>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-7DHI07TE
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-0A
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::6491:ee4a:b1e9:38f5%10(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 889847847
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-80-C6
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 1E-CE-51-F4-2D-52
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:
```

```

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCI
e Adapter
    Physical Address. . . . . : 1C-CE-51-F4-2D-52
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2404:7c00:48:6c03:dd5c:7b72:8ea2:de0d
(Preferred)
    Temporary IPv6 Address. . . . . : 2404:7c00:48:6c03:64f1:c57a:a3da:eed3
(Preferred)
    Link-local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5(Preferred
)
    IPv4 Address. . . . . : 192.168.101.5(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Thursday, September 10, 2026 2:27:47
PM
    Lease Expires . . . . . : Friday, September 11, 2026 2:27:46 PM
    Default Gateway . . . . . : fe80::72a5:6aff:feb3:259a%5
    192.168.101.1
    DHCP Server . . . . . : 192.168.101.1
    DHCPv6 IAID . . . . . : 85773905
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-8
0-C6
    DNS Servers . . . . . : fe80::1%5
    110.44.112.245
    110.44.113.245
    fe80::1%5
    NetBIOS over Tcpip. . . . . : Enabled

```

➤ arp -a

```

C:\Users\HP>arp -a

Interface: 192.168.101.5 --- 0x5
    Internet Address      Physical Address      Type
    192.168.101.1         70-a5-6a-b3-25-9a    dynamic
    192.168.101.6         22-3d-e3-ea-ac-eb    dynamic
    192.168.101.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    224.0.1.60            01-00-5e-00-01-3c    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0xa
    Internet Address      Physical Address      Type
    192.168.56.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    224.0.1.60            01-00-5e-00-01-3c    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

```